

Haoxue Ran

Department of Environmental and Municipal Engineering

Xi'an University of Architecture and Technology

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EDUCATION

Xi'an University of Architecture and Technology

Sep 2021 – Expected Jun 2027

Ph.D., Environmental Engineering (*Direct-Track, “Cuiying” Excellence Program*)

Advisor: Prof. Rui Miao

Guilin University of Technology

Sep 2017 – Jun 2021

B.E., Environmental Engineering (*Elite Engineering Program*)

Academic Rank: 1/30 within the program

RESEARCH EXPERIENCE

Graduate Student Researcher

Xi'an University of Architecture and Technology

Advisor: Prof. Rui Miao

2022 – Present

Department of Environmental and Municipal Engineering

- Developed a trace-dose Fenton pretreatment technology at neutral pH to mitigate membrane organic fouling, and elucidated its underlying mechanisms.
- Established a systematic analytical framework based on data-driven models to predict membrane fouling behavior and identify its key governing factors.
- Quantitatively decoupled the contributions of individual causal pathways to membrane fouling mitigation by integrating causal inference with micro-interfacial analysis techniques.
- Investigated the different mechanisms of divalent metal ions (Ca^{2+} , Mg^{2+}) on the efficacy of Fenton pretreatment in membrane organic fouling mitigation.
- Quantitatively decoupled the key factors and causal relationships of $\text{Ca}^{2+}/\text{Mg}^{2+}$ ions governing Fenton pretreatment for membrane fouling mitigation by integrating artificial intelligence with molecular dynamics simulations.

PUBLICATIONS & PATENTS

Journal Papers

1. **Ran, H.**, Miao, R.^{*}, Zhao, J.^{*}, Zhang, H., Wang, S., Zhang, J., Wang, P., Wang, L. and Yin, J. **2026**. Quantitative decoupling of the key factors and mechanisms of Fenton-pretreated membrane fouling mitigation via machine learning-assisted causal inference. *Water Research*, 303, 126247.
2. **Ran, H.**, Wang, Y., Miao, R.^{*}, Zhao, J.^{*}, Wang, S., Zhang, J., Wang, P., Wang, L. and Yin, J. **2026**. Effects of feature selection on predictive performance and interpretability in data-driven modeling:

A case study of membrane flux prediction after Fenton pretreatment. *Separation and Purification Technology*, 393, 137304.

3. Miao, R. *, **Ran, H.**, Yang, Y., Li, Y., Ma, Z., Lv, Y., Meng, X., He, M. and Wang, L. **2024**. In situ acid production by organic matter induced with trace homogeneous Fenton reagent for membrane fouling control. *Water Research*, 258, 121752.
4. Miao, R. *, Wang, Y., **Ran, H.**, Li, Y., Yang, C. and Wang, L. **2023**. Relative importance of drag force and adhesion forces in the fouling of ultrafiltration membranes by secondary effluent organic matter after pre-ozonation. *Journal of Membrane Science*, 678, 121677.

Patents & Software Copyrights

- Miao, R., **Ran, H.**, Wang, L., Lv, Y., Yang, C. and Wang, Y. (**2023**). *Method and application for membrane fouling control via Fe^{2+} - H_2O_2 induced in situ acid production from organic pollutants*. Chinese Invention Patent, Application No: CN202310381227.4.
- Miao, R., **Ran, H.**, Zhao, C., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Method for membrane fouling mechanism identification and regulation based on causal inference*. Chinese Invention Patent, Application No: CN202511264800.9.
- Miao, R., **Ran, H.**, Yang, Y., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Machine learning-based method and system for the optimal control of membrane fouling pretreatment*. Chinese Invention Patent, Application No: CN202510539314.7.
- Miao, R., **Ran, H.**, Wang, Y., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Method and system for membrane fouling prediction and key factor identification/regulation in Fenton pretreatment*. Chinese Invention Patent, Application No: CN202511401234.1.
- Bai, S., **Ran, H.**, Wu, H. and Zhou, P. (**2020**). *Intelligent maintenance device for green building vegetation*. Chinese Utility Model Patent, Patent No: ZL202020307040.1.
- **Ran, H.**, Miao, R., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Dosage optimization system for Fenton pretreatment-based membrane fouling mitigation under various operating conditions*. Chinese Computer Software Copyright, Registration No: 2025SR2214520.

RESEARCH PROJECTS & GRANT PROPOSALS

Natural Science Basic Research Program of Shaanxi Province (Youth Project, Class A)

Research on theory and application of membrane fouling control based on multidimensional influencing factors.

Role: Co-drafted proposal.

2026

National Key Research and Development Program of China (Sub-project)

Interfacial migration mechanisms of salt ions in brine and fouling prevention/control technologies for membranes/adsorbents.

Role: Graduate Research Assistant.

2022.10 – 2026.09

National Natural Science Foundation of China (NSFC, General Program)

Mechanism and regulation of hydration repulsion for membrane fouling mitigation during ultrafiltration.

Role: Graduate Research Assistant.

2021.01 – 2024.12

Youth Innovation Team of Shaanxi Universities

Innovative Team of Membrane Separation Technology for Valuable Resource Extraction from Salt Lake Brine.

Role: Graduate Research Assistant.

2023.01 – 2026.12

International Science and Technology Cooperation Project of Shaanxi Province

High-efficiency reuse process of municipal secondary effluent centered on ultrafiltration technology.

Role: Graduate Research Assistant.

2022.01 – 2023.12

National Natural Science Foundation of China (NSFC, General Program)

General Program: Self-cleaning mechanism and regulation of microorganisms on MBR membrane surface induced by self-forming dynamic membranes.

Role: Research Contributor.

2024.01 – 2027.12

Enterprise-funded Collaborative R&D Projects

Resource utilization of kitchen and septic tank domestic wastes in Shendong Coal.

Role: Research Contributor.

2022.07 – 2026.12

Recycled utilization of mine water for promoting low-carbon emission reduction in boilers.

Role: Research Contributor.

2025.05 – 2025.12

HONORS & AWARDS

Research & Innovation Awards

- Third Prize, 19th “Challenge Cup” National Undergraduate Academic Science and Technology Competition 2025
- Best Presentation Award, 20th National Graduate Environmental Forum (Tongji University) 2025
- Outstanding Undergraduate Thesis Award, Guilin University of Technology 2021
- Third Prize, 16th “Challenge Cup” National Undergraduate Academic Science and Technology Competition 2019
- Grand Prize, “Challenge Cup” National Undergraduate Academic Science and Technology Competition (*Guangxi Regional Selection*) 2019

Scholarships & Honors

- “Yu Dan” Corporate Excellence Scholarship, Xi’an University of Architecture and Technology 2025
- Second-class Scholarship of Xi’an University of Architecture and Technology 2025
- Second-class Scholarship of Xi’an University of Architecture and Technology 2024
- First-class Scholarship of Xi’an University of Architecture and Technology 2023

- Second-class Scholarship of Xi'an University of Architecture and Technology 2022
- Outstanding Graduate Honor, Guangxi Autonomous Region, China 2021
- Guangxi Provincial Government Scholarship 2020
- Special-class Scholarship of Guilin University of Technology 2020
- Outstanding Student Award, Guilin University of Technology 2020
- First-class Scholarship of Guilin University of Technology 2019
- Outstanding Student Award, Guilin University of Technology 2019
- First-class Scholarship of Guilin University of Technology 2018
- Elite Engineer Fellowship, Guilin University of Technology 2018
- Outstanding Student Award, Guilin University of Technology 2018

RESEARCH PRESENTATIONS

1. "AI-driven Analysis of Membrane Fouling Mechanisms," *8th IWA Regional Membrane Technology Conference (IWA-RMTC 2026)*, Shanghai, China, Aug 2026. (Oral Presentation).
2. "Deciphering Complex Membrane Fouling Mechanisms via Explainable Machine Learning: A Case Study of Fenton-Pretreated Membranes," *Academic Exchange on Green Low-Carbon and Sustainable Development (IEEPA Annual Meeting)*, Hangzhou, China, Dec 2025. (Oral Presentation).
3. "Mechanisms of Trace $\text{Fe}^{2+}/\text{H}_2\text{O}_2$ Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *2025 China Membrane Industry Intelligent Manufacturing and Application Technology Conference*, Xi'an, China, Nov 2025. (Oral Presentation).
4. "Mechanisms of Trace $\text{Fe}^{2+}/\text{H}_2\text{O}_2$ Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *20th National Graduate Environmental Forum (Tongji University)*, Shanghai, China, May 2025. (Oral Presentation, **Best Presentation Award**).
5. "Mechanisms of Trace $\text{Fe}^{2+}/\text{H}_2\text{O}_2$ Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *19th National Environmental Ph.D. Academic Conference (Tsinghua University)*, Beijing, China, May 2025. (Oral Presentation & **Session Chair**).

TEACHING AND MENTORING EXPERIENCE

Graduate Teaching Assistant **Xi'an University of Architecture and Technology**
 Course: *Membrane Separation Theory and Technology* (Instructor: Prof. Rui Miao) **Spring 2022**
 • Assisted in course logistics, graded homework assignments, and provided feedback to the instructor (about 120 students).

Research Seminar Coordinator **Miao Research Group**
 Shaanxi Membrane Separation Research Institute **2021 – Present**

- Scheduled academic topics and managed logistics for monthly large-scale research group seminars (about 20–30 participants).

Mentoring and Research Advising

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| • Jiayi Zhang (Undergraduate Researcher) | 2022 – Present |
| • Yifan Yang (M.S. Student) | 2022 – 2025 |
| • Siyu Wang (M.S. Student) | 2023 – 2026 |
| • Yuqing Lei (M.S. Student) | 2024 – Present |
| • Fangru Zhu (M.S. Student) | 2025 – Present |

PROFESSIONAL SERVICE AND LEADERSHIP

Assisted Review: *Water Research, Journal of Membrane Science, Chemical Engineering Journal, Desalination, Separation and Purification Technology, Journal of Water Process Engineering.*

Student Laboratory Manager: Miao Research Group, Shaanxi Membrane Separation Research Institute.

Safety & Operations Coordinator: Department of Environmental and Municipal Engineering, Xi'an University of Architecture and Technology.

RESEARCH SKILLS

Analytical Characterization: Liquid Chromatography-Time of Flight Mass Spectrometry (LC-TOF-MS), Quartz Crystal Microbalance with Dissipation (QCM-D), Confocal Laser Scanning Microscopy (CLSM), Liquid Chromatography with Organic Carbon Detection (LC-OCD), Scanning Electron Microscopy (SEM), Inductively Coupled Plasma Mass Spectrometry (ICP-MS), Atomic Force Microscopy (AFM), Fourier Transform Infrared Spectroscopy (FTIR), Zeta Potential & Hydrodynamic Diameter Analysis, Laser Diffraction Particle Size Analysis.

Modeling & Computation: Data-driven predictive modeling (CatBoost, TabPFN, Random Forest, Deep Learning), Explainable Artificial Intelligence (XAI), Causal Inference & Directed Acyclic Graphs (DAGs), Molecular Dynamics (MD) Simulations.